



Swiss Re

Breast and gynaecological cancer

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Health Claims Forum 27 November 2014

Agenda

- Breast cancer
- Gynaecological cancer
 - ovary
 - uterine body (endometrium)
 - uterine cervix
- Diagnosis, staging and treatment

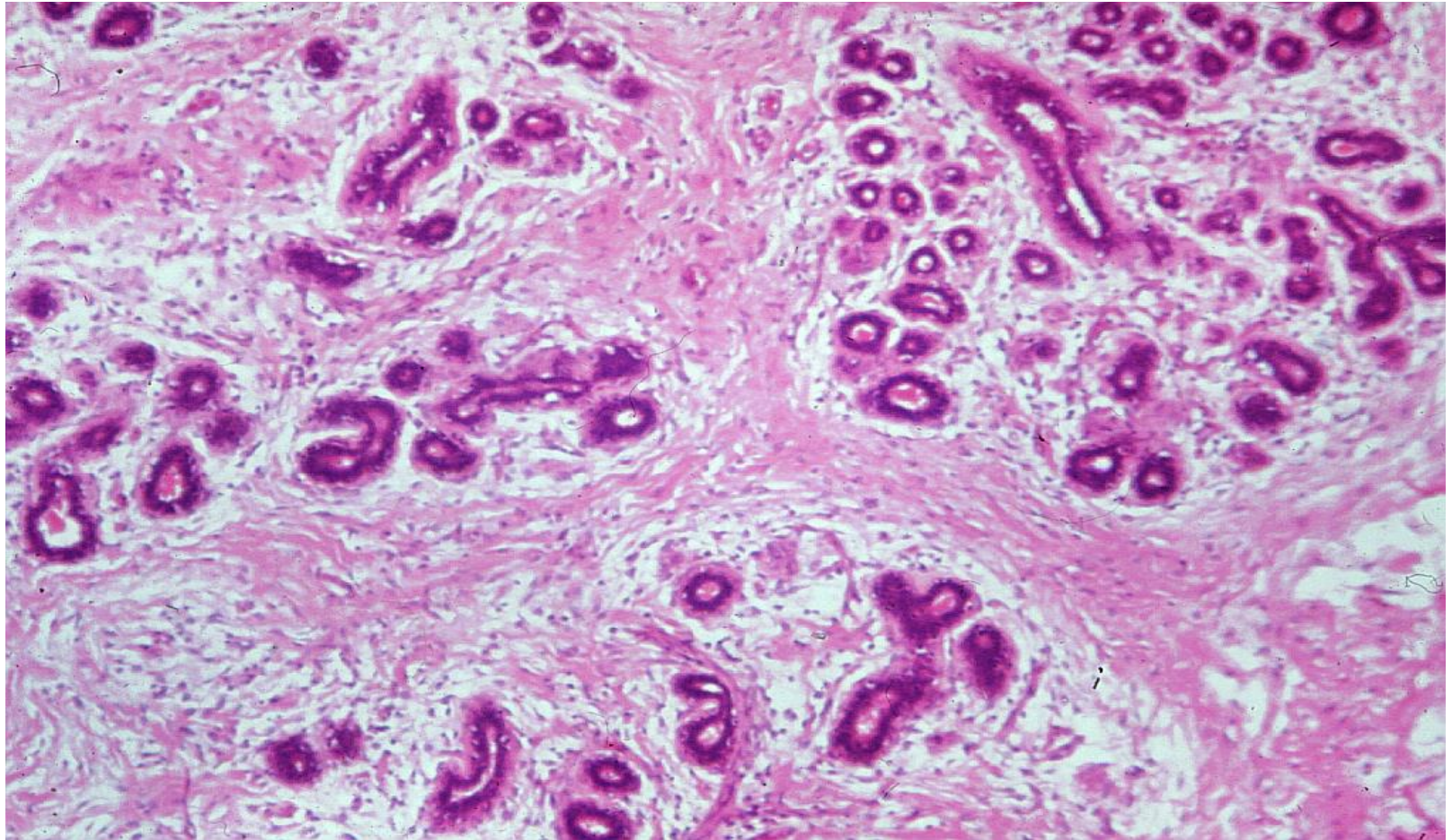
Breast cancer incidence UK

- New patients per annum 25,000
- Deaths per annum 15,000
- Life-time risk (female) 1:12
- Risk by age 50 (female) 1:70

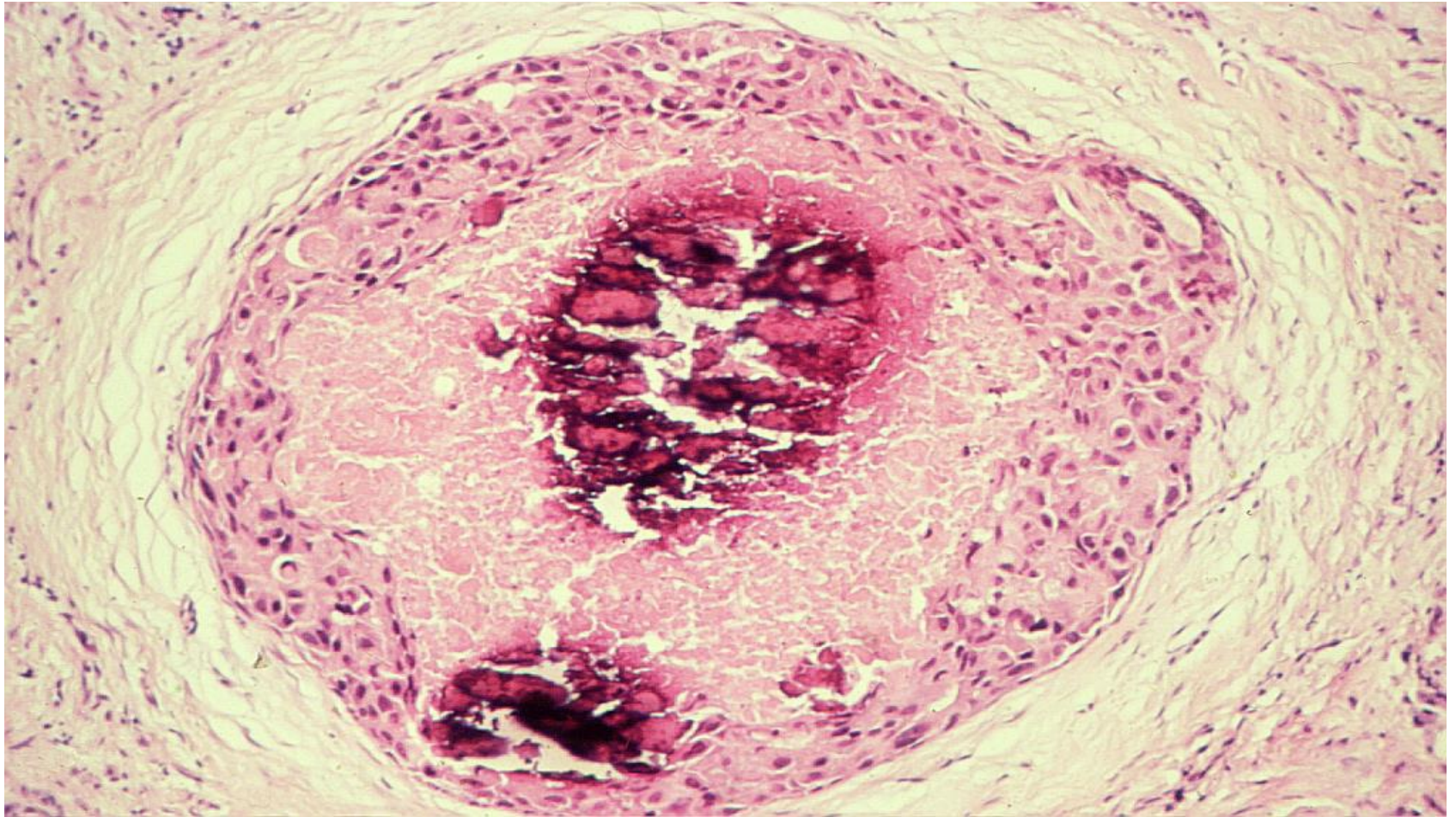
Primary assessment of breast cancer

- Biopsy - Histology
- Clinical examination
- Investigations
 - Haematological
 - Biochemical
 - Imaging (mammography, ultra sound, MRI, bone scan)

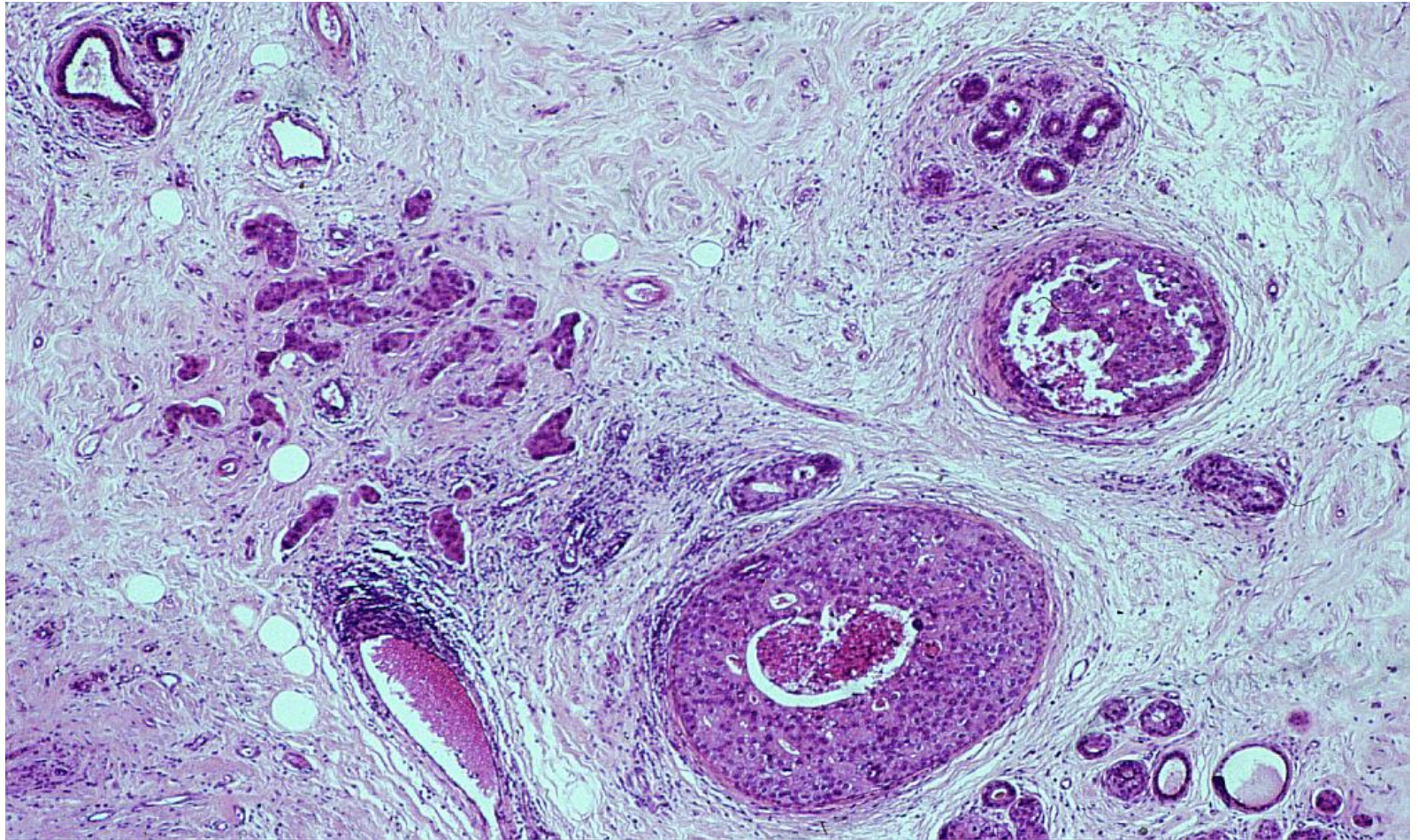
Normal breast



Ductal carcinoma *in situ* (DCIS)



DCIS with invasion



DCIS - mortality

- Mortality in women treated for DCIS is low
- SEER database: n = 5547 patients diagnosed between 1984 and 1989,
- 0.7% had died from breast cancer within 5 years and 1.9% by 10 years.
- Relative to the general population, this represented an SMR of 1.9 (95% CI 1.5 – 2.3).
- However, these women were significantly less likely to die from either cardiovascular disease (SMR 0.6; 95% CI 0.5 – 0.7) or all causes combined (SMR 0.8; 95% CI 0.7 – 0.8).

Ernster VL et al. Arch Intern Med 2000; 160: 953

Breast cancer: determinants of outcome

- Tumour type
- Histological differentiation
- Staging
- Treatment

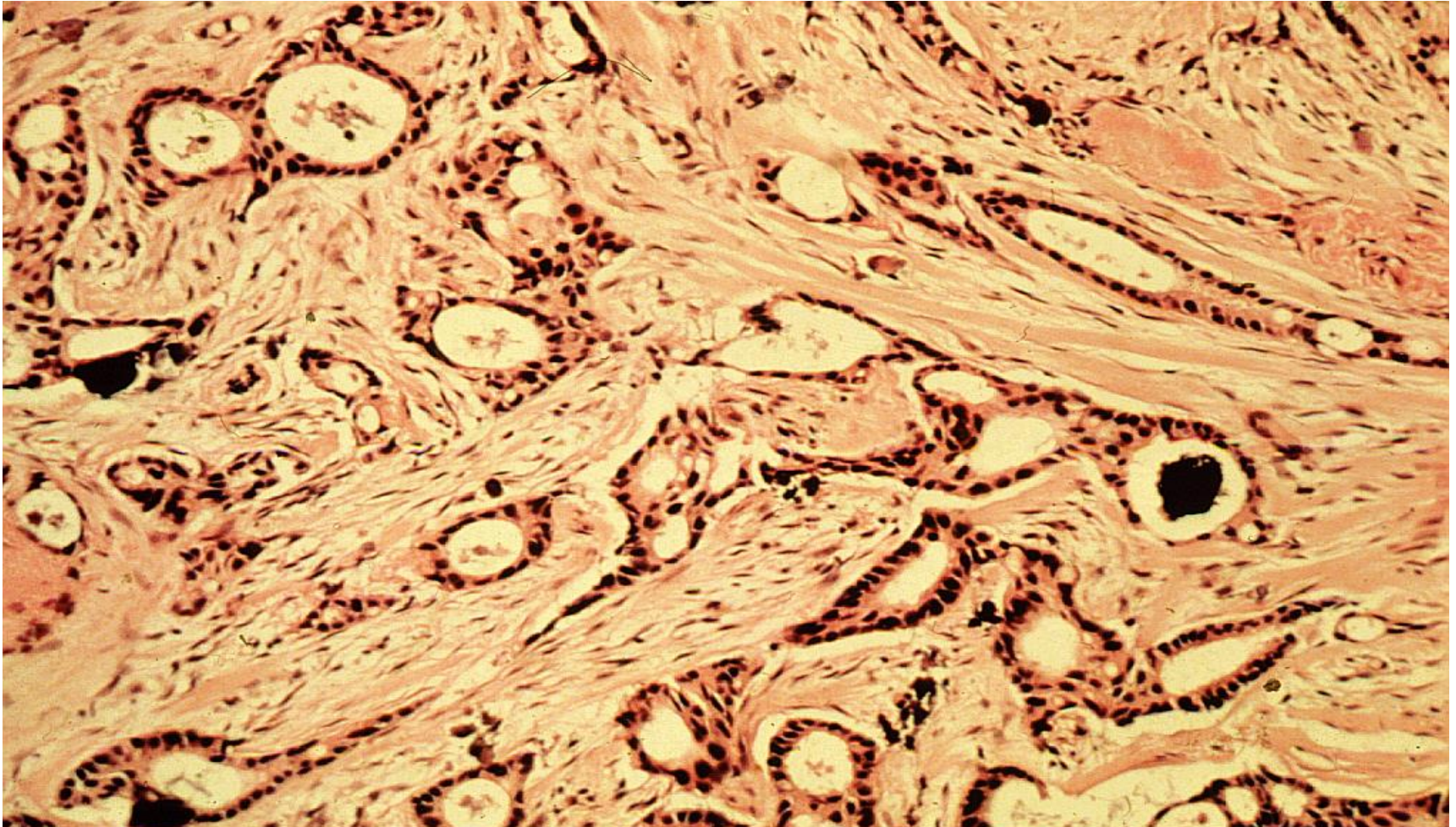
Breast cancer types

- Ductal
- Lobular
- Tubular
- Muroid
- Medullary

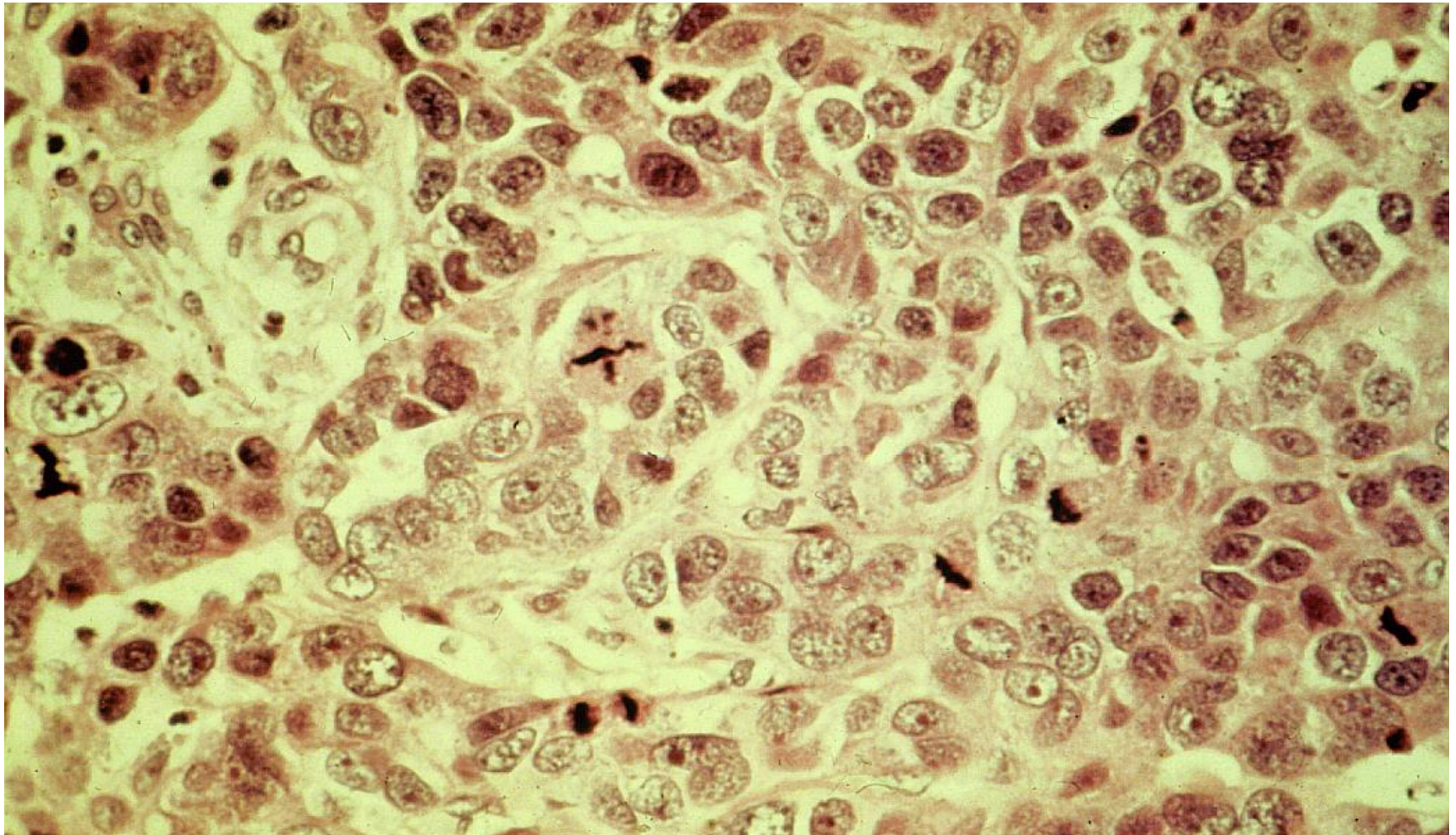
Breast cancer differentiation

- Well-differentiated
- Poorly differentiated = Anaplastic

Breast carcinoma: well-differentiated



Breast carcinoma: poorly-differentiated



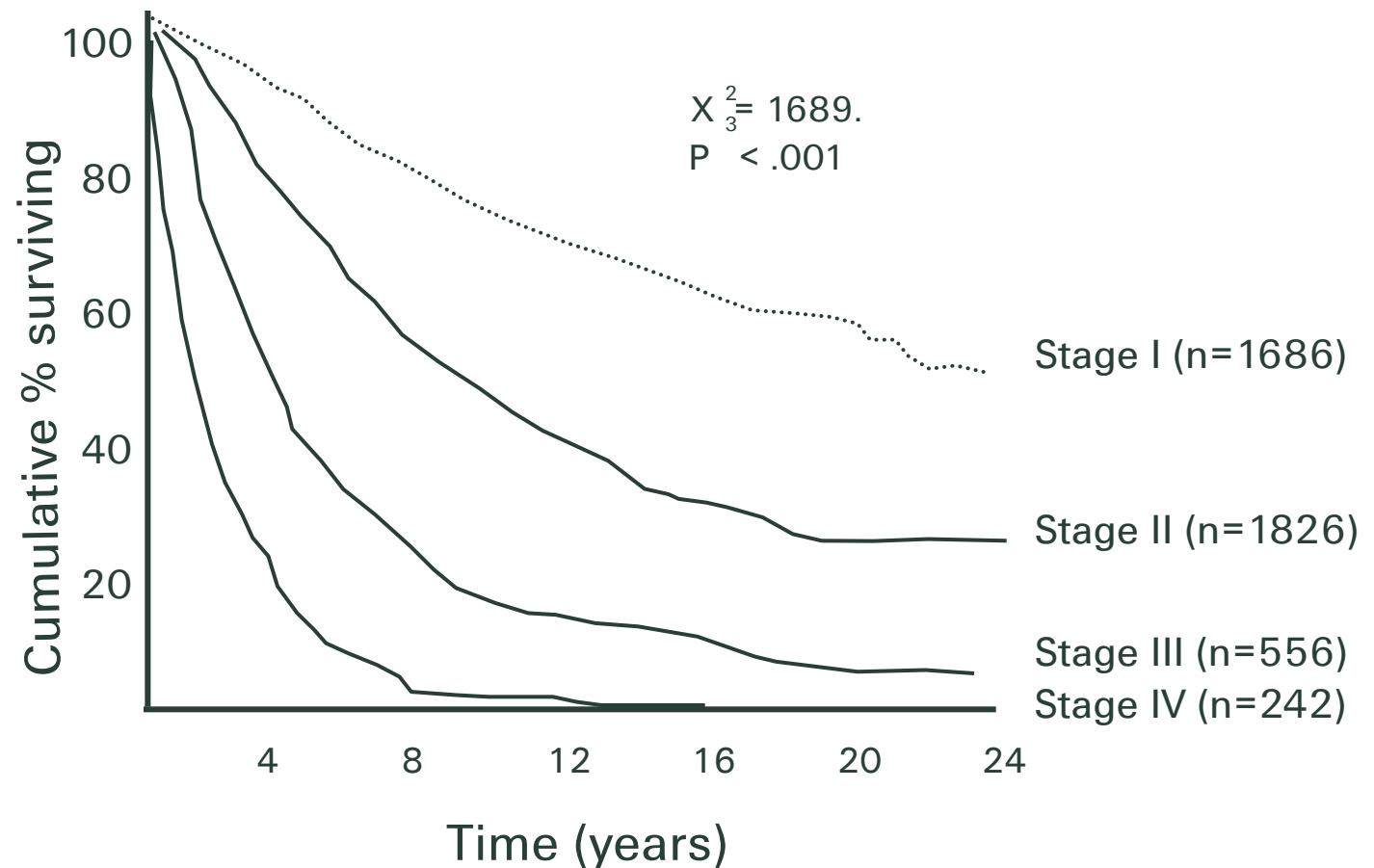
Staging of Breast Cancer

Stage

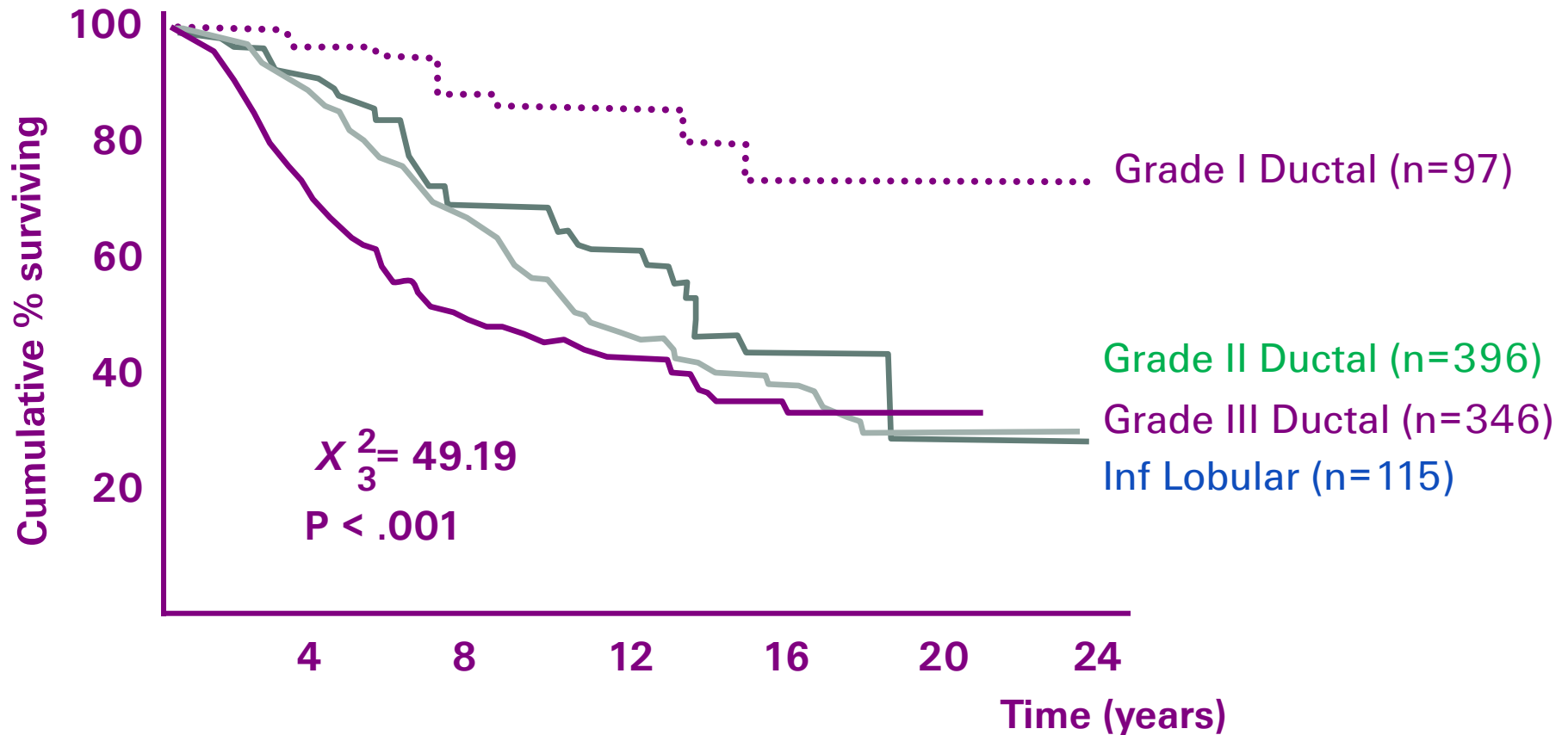
0	In situ, no invasion
1	Invasive, confined to breast
2	Involvement of axillary lymph nodes
3	Locally advanced disease
4	Distant metastases

Survival: breast cancer

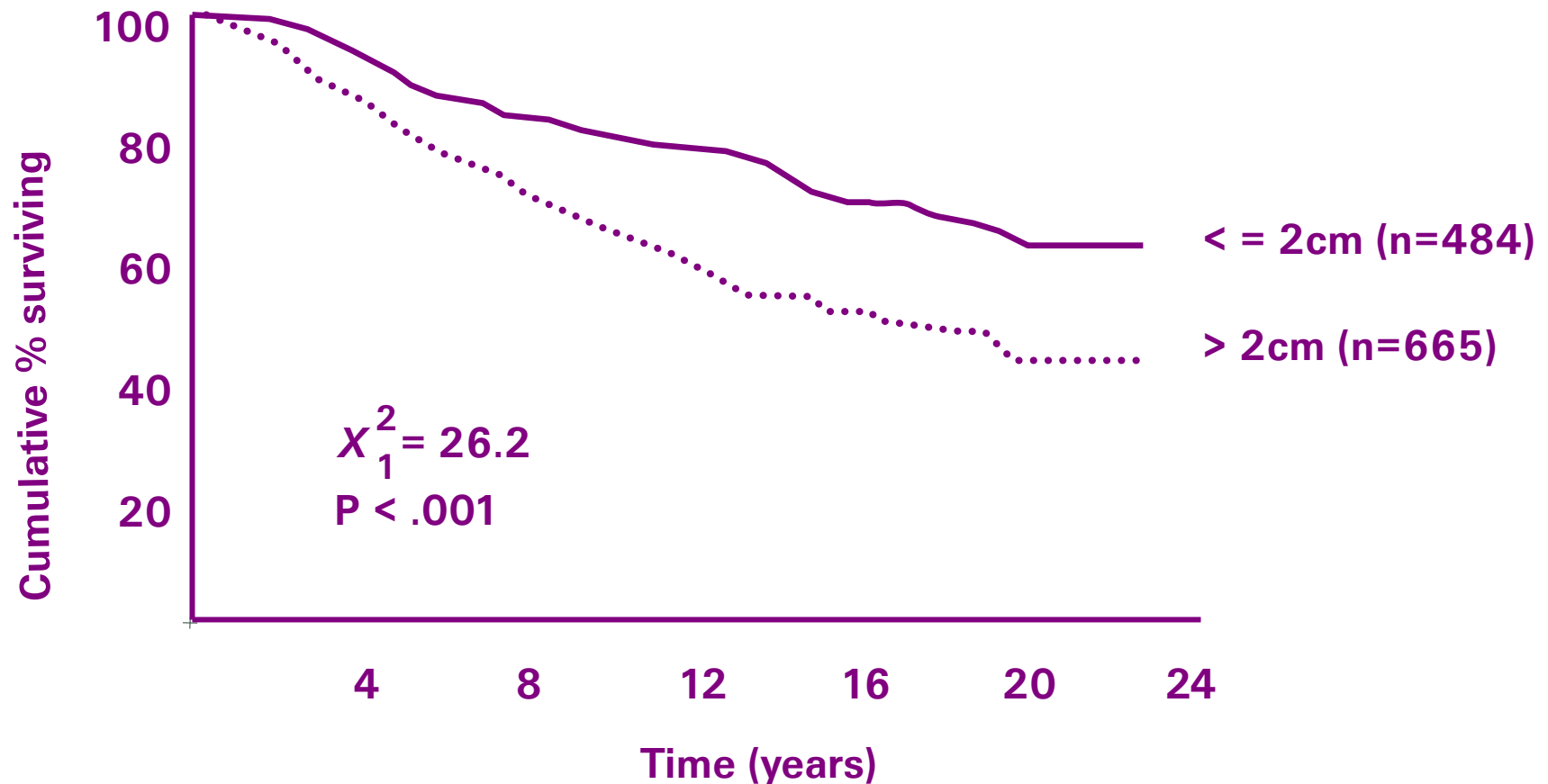
Guy's Hospital Breast Unit (1975 - 1999)



Survival: breast cancer - nodes positive 1-3 Guy's Hospital Breast Unit (1975-1999)



Survival: breast cancer - node negative,
not grade I, Guy's Hospital Breast Unit
(1975-1999)



Treatment of breast cancer

- Surgery
- Radiotherapy
- Systemic
 - Endocrine
 - Cytotoxic drugs
 - Antibodies (trastuzumab, Herceptin)

Breast cancer: mortality reductions from adjuvant systemic therapy

- *Adjuvant! Online* is based on SEER data and calculates risk reductions from the use of adjuvant systemic therapy determined from the statistical overviews of the Early Breast Cancer Trialists' Collaborative Group (*EBCTCG 1998a/b; Ravdin PM et al, 2001*).
- For a given age, state of general health and prognostic category of breast cancer, *Adjuvant! Online* gives the following proportions of patients at 10 years after primary treatment:
 - % alive
 - % dead from breast cancer
 - % dead from other causes
- For the proportion dying of breast cancer, the model estimates the reductions in deaths from the use of adjuvant endocrine therapy and/or chemotherapy.

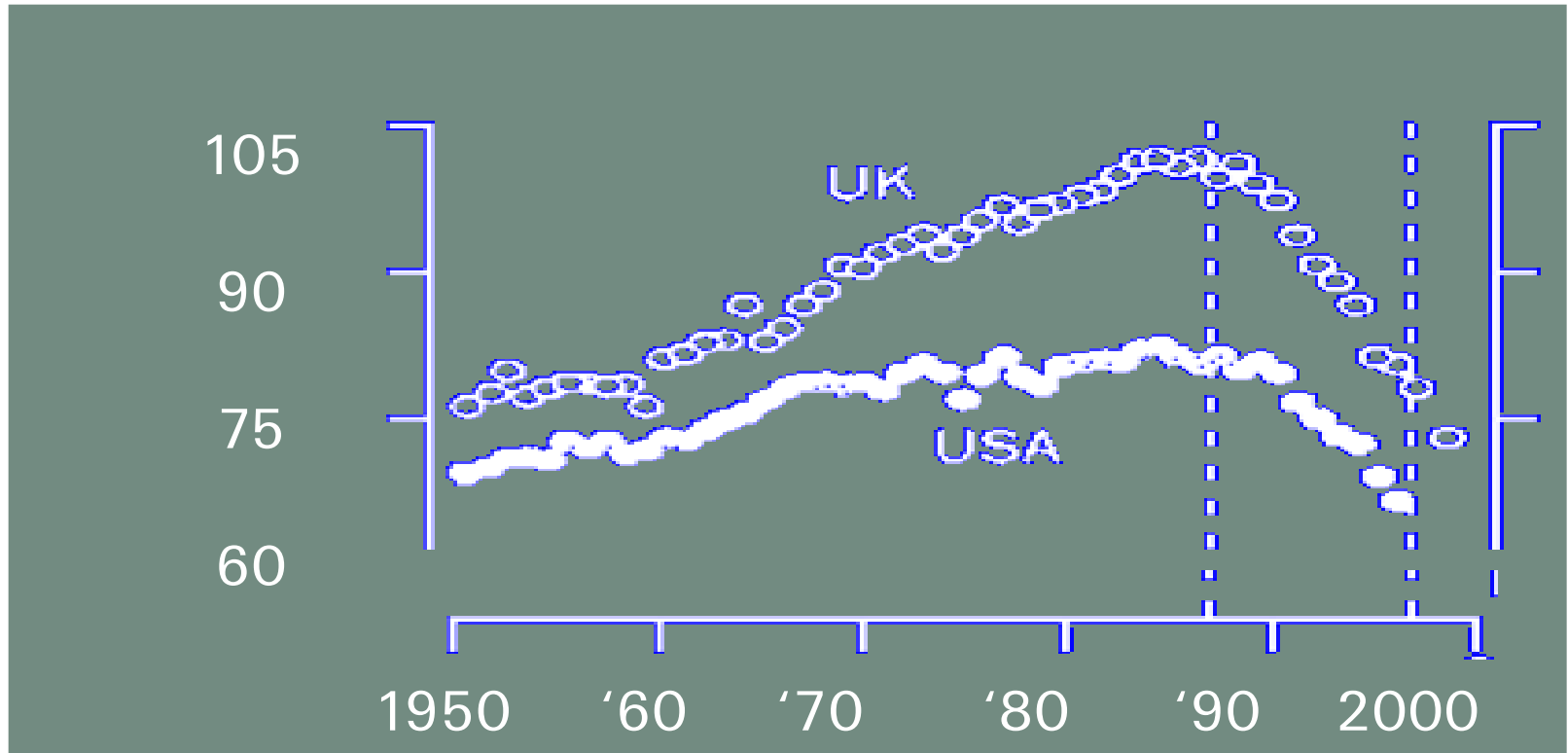
Breast tumours: example of EBR data

- Extra deaths from breast cancer per thousand women at 10 years after treatment for primary operable disease by age and ER status for *node negative, grade 2, 3.1 – 5.0 cm*

Age	30	40	50	60	70
ER- surgery only	329	328	325	318	303
ER+ surgery only	339	238	235	227	210
ER+ with endocrine therapy	244	164	167	163	157

<https://www.adjuvantonline.com/online>

Breast cancer: annual death rate per 100,000 women



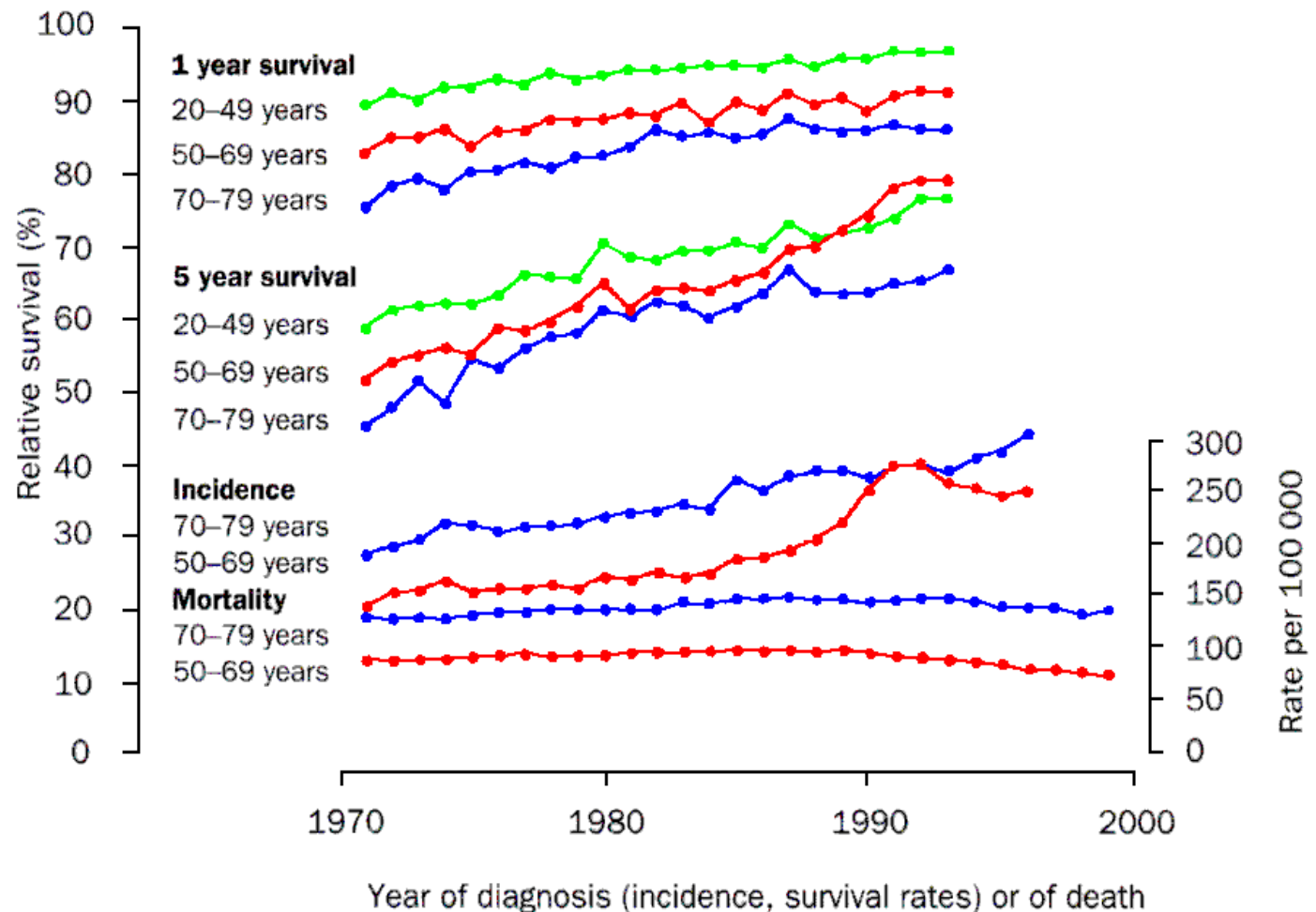
Lancet 1998: 355; 1822

Breast cancer: 5-year relative survival by year of diagnosis

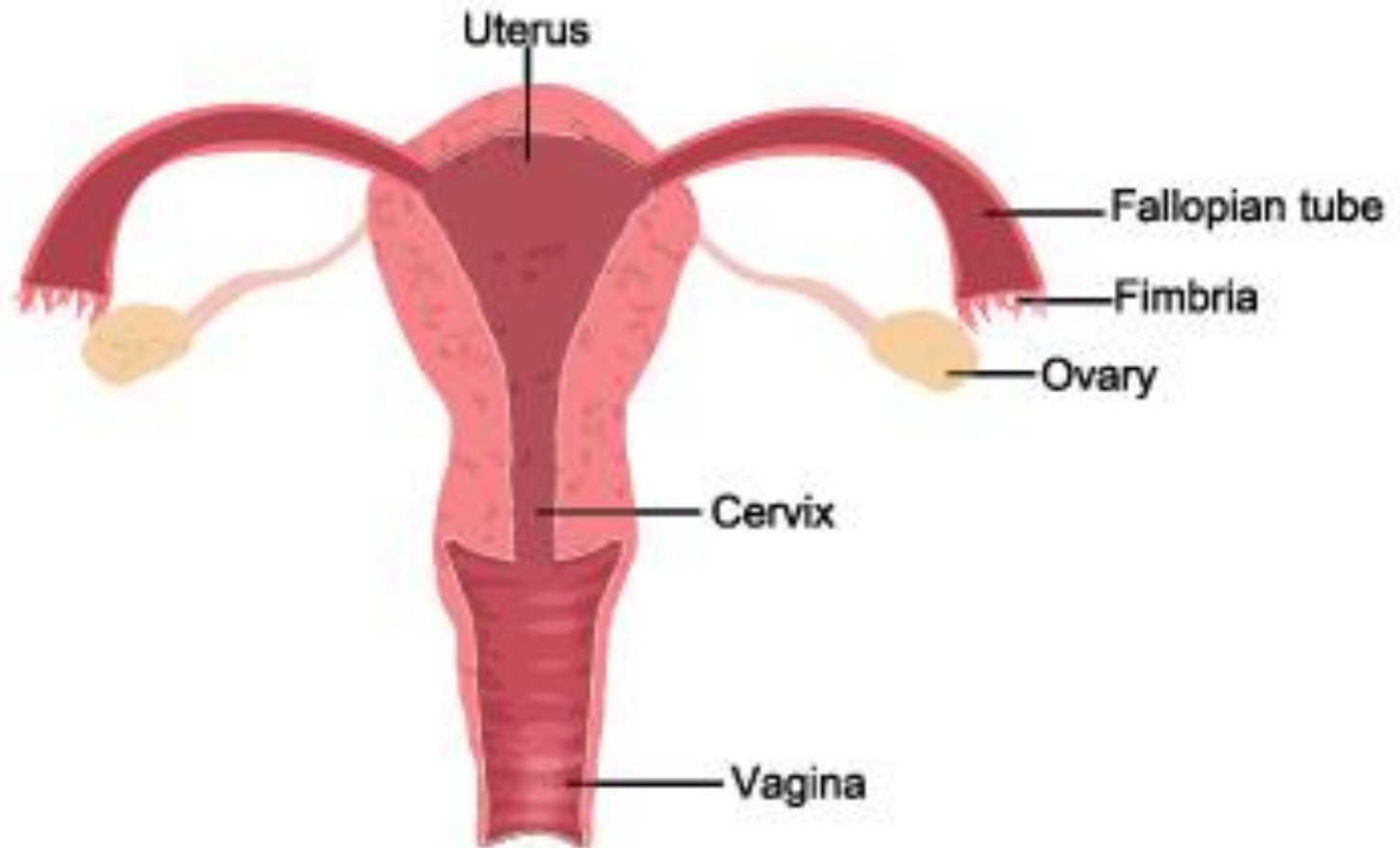
1970	1980	1990	2000
68%	77%	85%	91%

Source: <http://seer.cancer.gov>

Breast cancer: trends in incidence, survival & mortality - UK



Female reproductive system



Gynaecological tumours

- Issues in Claims
 - Borderline ovarian tumours of low malignant potential
 - Stage 1 endometrial carcinoma
 - Cervical intraepithelial neoplasia (CIN)

Cancer of the ovary

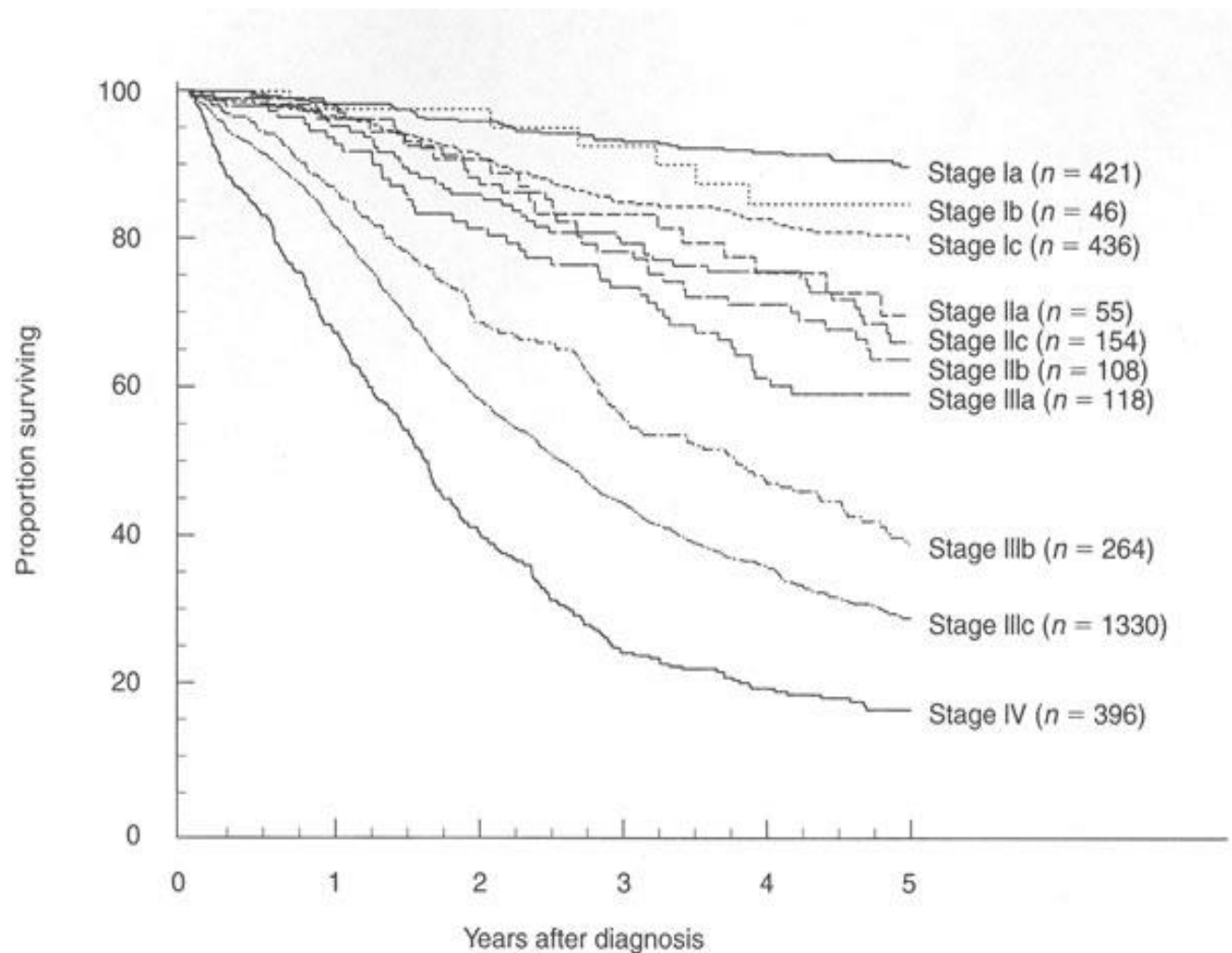
- 95% are carcinomas; other types are:
 - germ cell tumours
 - sex cord-stromal tumours
- 10% are inherited (BRCA1/2 mutations)

Staging of ovarian carcinoma:

Federation Internationale de Gynecologie et d'Obstetrique (FIGO)

- Stage 1 (growth limited to the ovaries)
 - Stage 1a - one ovary involved
 - Stage 1b - two ovaries involved
 - Stage 1c - one or both ovaries with ruptured capsule or ascites
- Stage 2 (pelvic extension)
 - Stage 2a - involvement of uterus and/or fallopian tubes
 - Stage 2b - involvement of other pelvic tissues
 - Stage 2c - pelvic extension with ruptured capsule or ascites
- Stage 3 (intra-abdominal extension outside the pelvis and/or involvement of retroperitoneal or inguinal lymph nodes)
- Stage 4 (distant metastases)

Carcinoma of the ovary - survival: patients treated 1993-5 (FIGO annual report 2002)



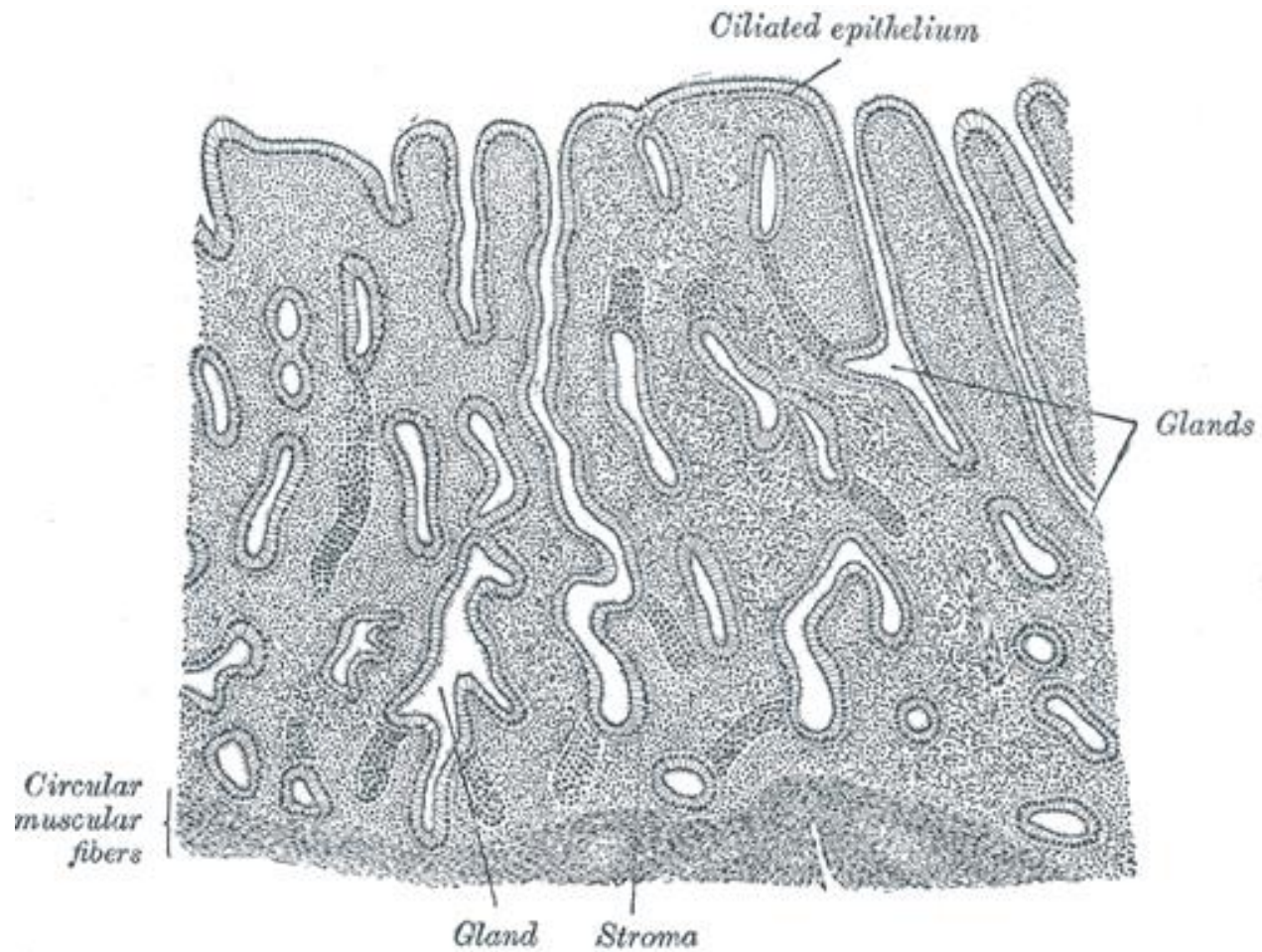
Ovarian carcinoma: 5-year survival by stage

Stage	Invasive cancer		Borderline tumours	
	Patients	% 5-yr survival	Patients	% 5-yr survival
Ia	421	89.9	296	95.6
Ib	46	84.7	28	95.9
Ic	436	80.0	90	96.3
IIa	55	69.9	6	100.0
IIb	108	63.7	7	85.7
IIc	154	66.5	14	59.5
IIIa	118	58.5	14	71.4
IIIb	264	39.9	22	62.0
IIIc	1330	28.7	25	45.0
IV	396	16.8	18	-

Cancer of the body of the uterus (corpus uteri)

- Arises in:
 - endometrium (common)
 - 80% are adenocarcinomas (endometrioid)
 - other types of carcinoma (clear cell, serous)
 - stromal sarcoma
 - myometrium (rare)
 - leiomyosarcoma

Endometrium



Endometrial cancer: - staging before 2010

Stage	Definition
1a	Tumour limited to endometrium
1b	Invades less than one half of myometrium
1c	Invades one half or more of myometrium
2a	Endocervical involvement only of the cervix
2b	Cervical stromal invasion
3a	Invades serosa of corpus uteri
3b	Vaginal metastases
3c	Metastases to pelvic and/or para-aortic lymph nodes
4a	Invasion of bladder and/or bowel
4b	Distant metastases

Endometrial cancer: - 5-year survival by stage

Stage	n	%5-year survival
1a	975	88.9
1b	2035	90.0
1c	986	80.7
2a	342	79.9
2b	367	72.3
3a	457	63.4
3b	101	38.8
3c	200	51.1
4a	57	19.9
4b	174	17.2

FIGO Annual Report, 2001

Endometrial carcinoma: stage 1 (confined to corpus uteri)

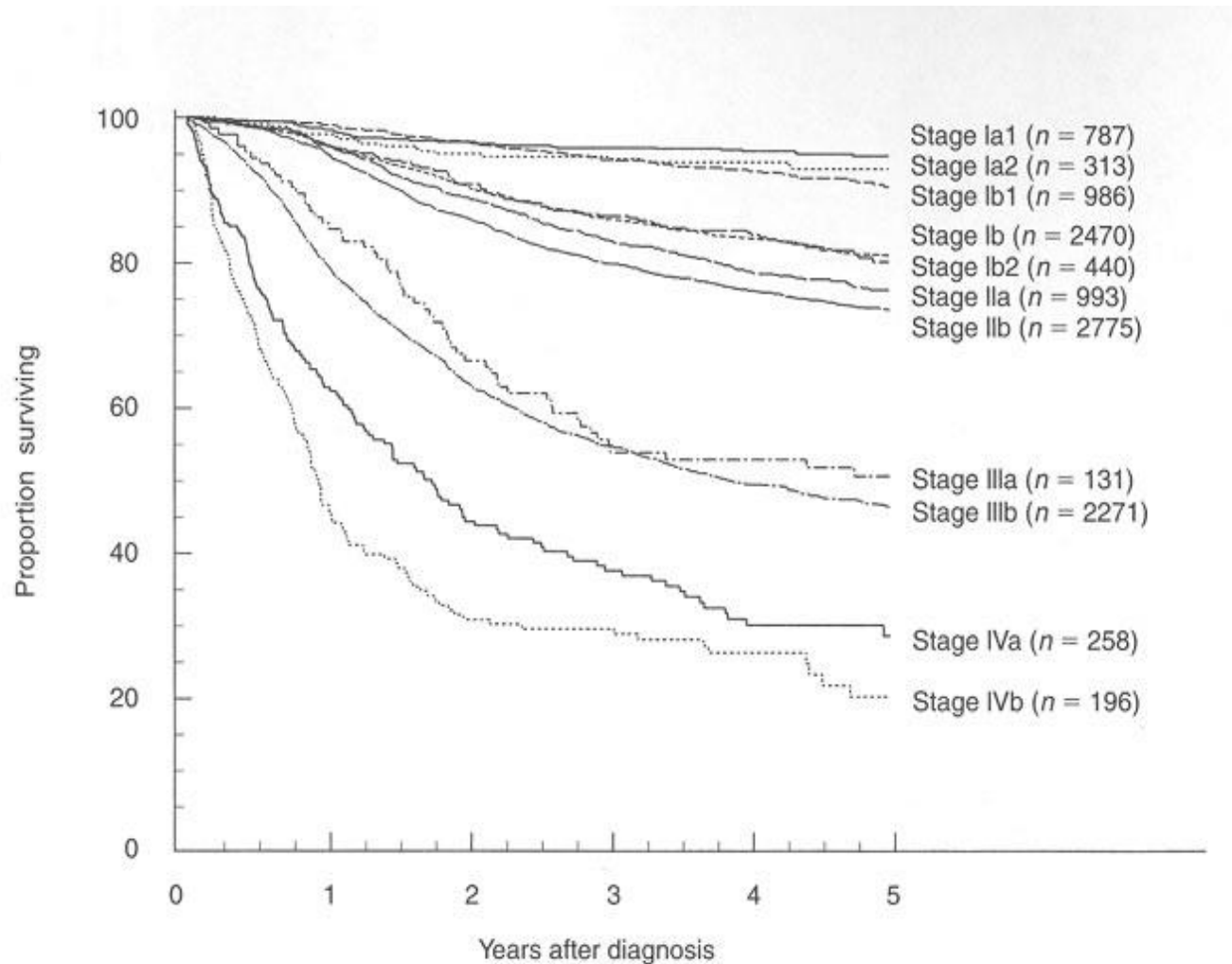
Stage	FIGO (AJCC, 2002)	FIGO (AJCC,2010)
1A	Limited to endometrium	Limited to endometrium or invades less than one-half of myometrium
1B	Invades less than one-half of myometrium	Invades one-half or more of myometrium
1C	Invades one-half or more of myometrium	-

In the absence of myometrial invasion, it may be difficult to distinguish between endometrial hyperplasia and carcinoma

Cancer of the uterine cervix

- 90% of cases are associated with human papilloma virus (HPV) infection
- 90% are squamous cell carcinomas; 10% adenocarcinoma
- Invasive cancer develops from pre-malignant changes over 10 to 20 years
- Prevented by cervical screening

Carcinoma of the cervix uteri - survival: patients treated 1993-5 (FIGO annual report 2002)



Cervical cytology

Papanicolaou's

Test

I

II

III

IV

V

Histology

Normal

Mild dysplasia

Moderate dysplasia

Severe dysplasia

Carcinoma *in situ*

Invasive carcinoma

FIGO ⁽¹⁾

and WHO ⁽²⁾

Normal

CIN I

CIN II

CIN III

Carcinoma

⁽¹⁾ Fédération Internationale de Gynécologie et d'Obstétrique

⁽²⁾ World Health Organisation

Cervical dysplasia: interobserver variation in histopathological grading

Pathologist	None	Mild	Moderate	Severe	cis
A	3%	24%	41%	28%	5%
B	5%	32%	32%	29%	2%
C	7%	25%	49%	18%	2%
D	2%	25%	29%	36%	9%